Everything Everywhere All at Once

Reimplementation of DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

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Background

Problem: In generating new images of a given subject, text-to-image diffusion models struggle to preserve the identity of a subject across different generated scenes

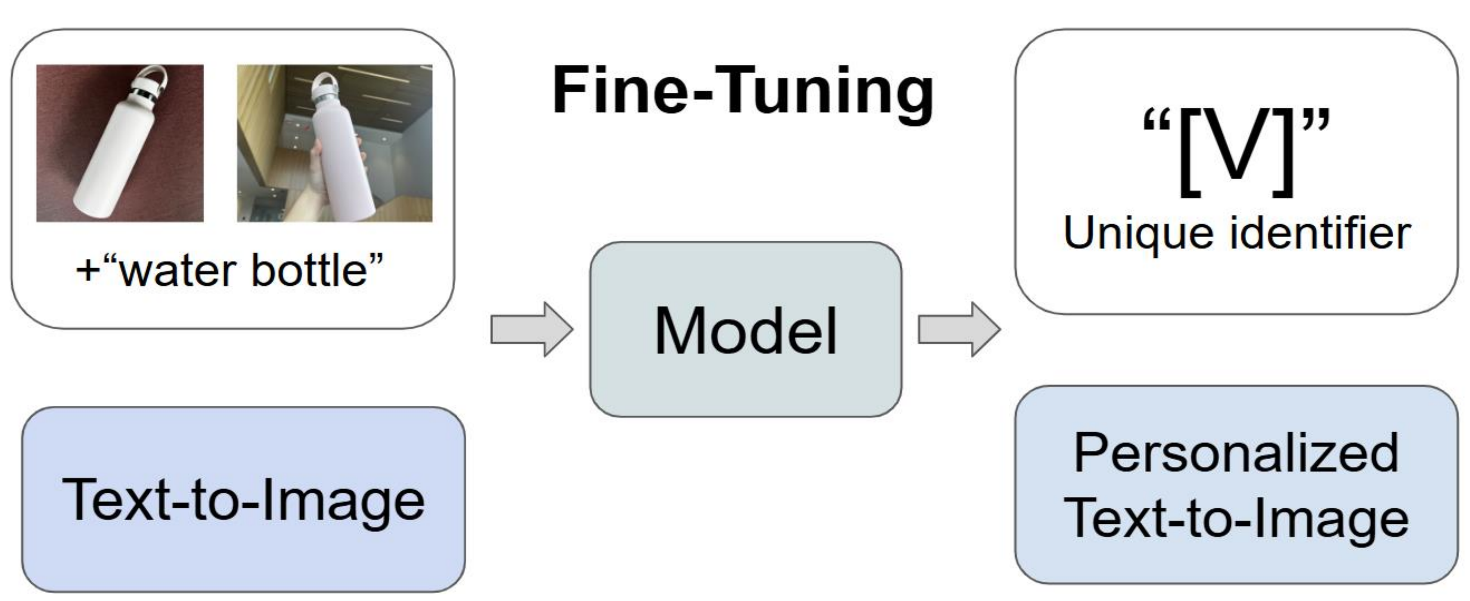
DreamBooth goal: Given 3-5 images, bind a rare identifier token to the subject to preserve subject fidelity through recontextualization, art renditions, etc.

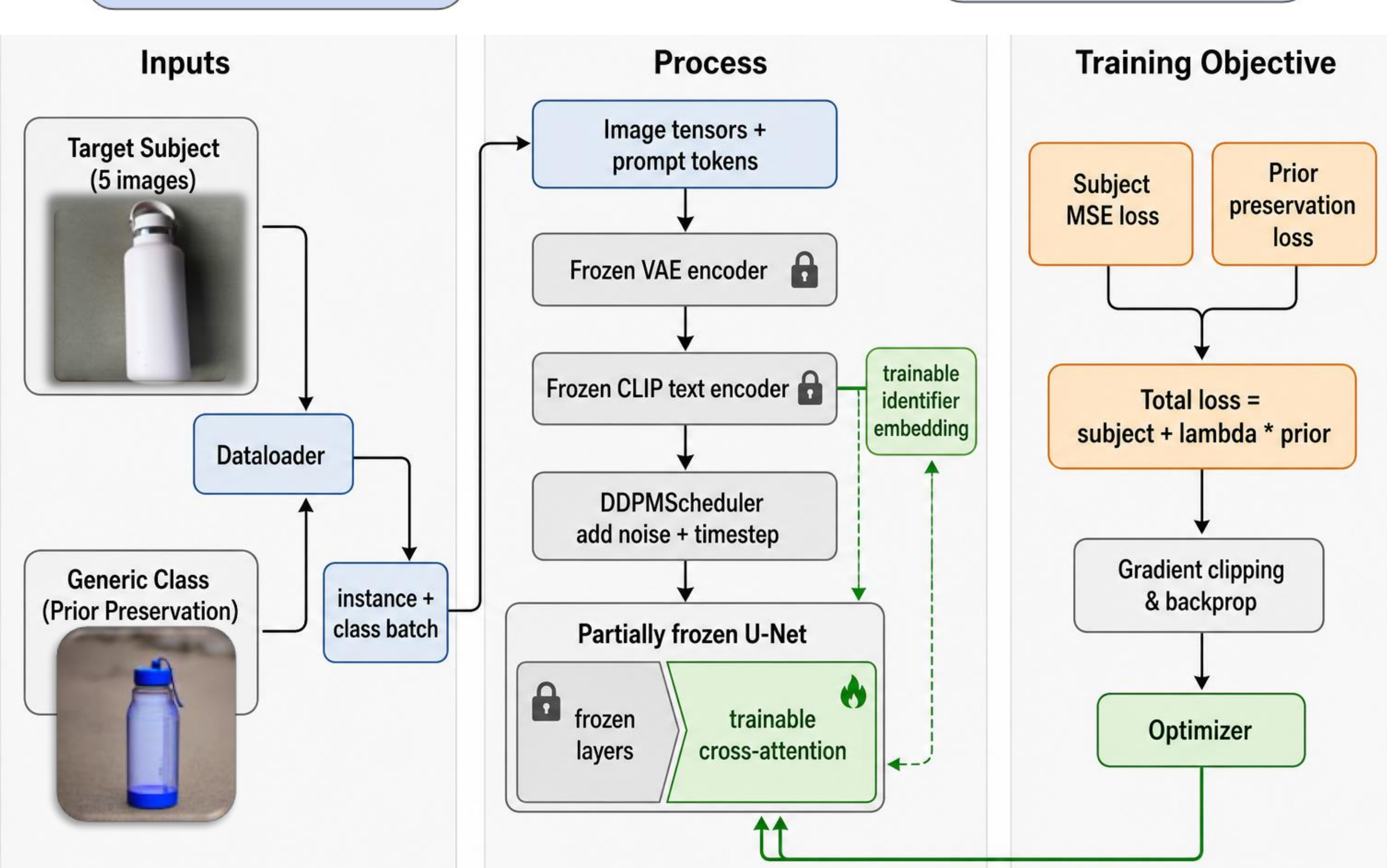
Our question: Can a partially frozen, low-memory DreamBooth variant still achieve similar useful subject binding?

Hypothesis

Low/no prior preservation may be enough for subject binding in our frozen model to achieve decent results with good DINO/CLIP-I scores at a much lower memory cost.

Methodology



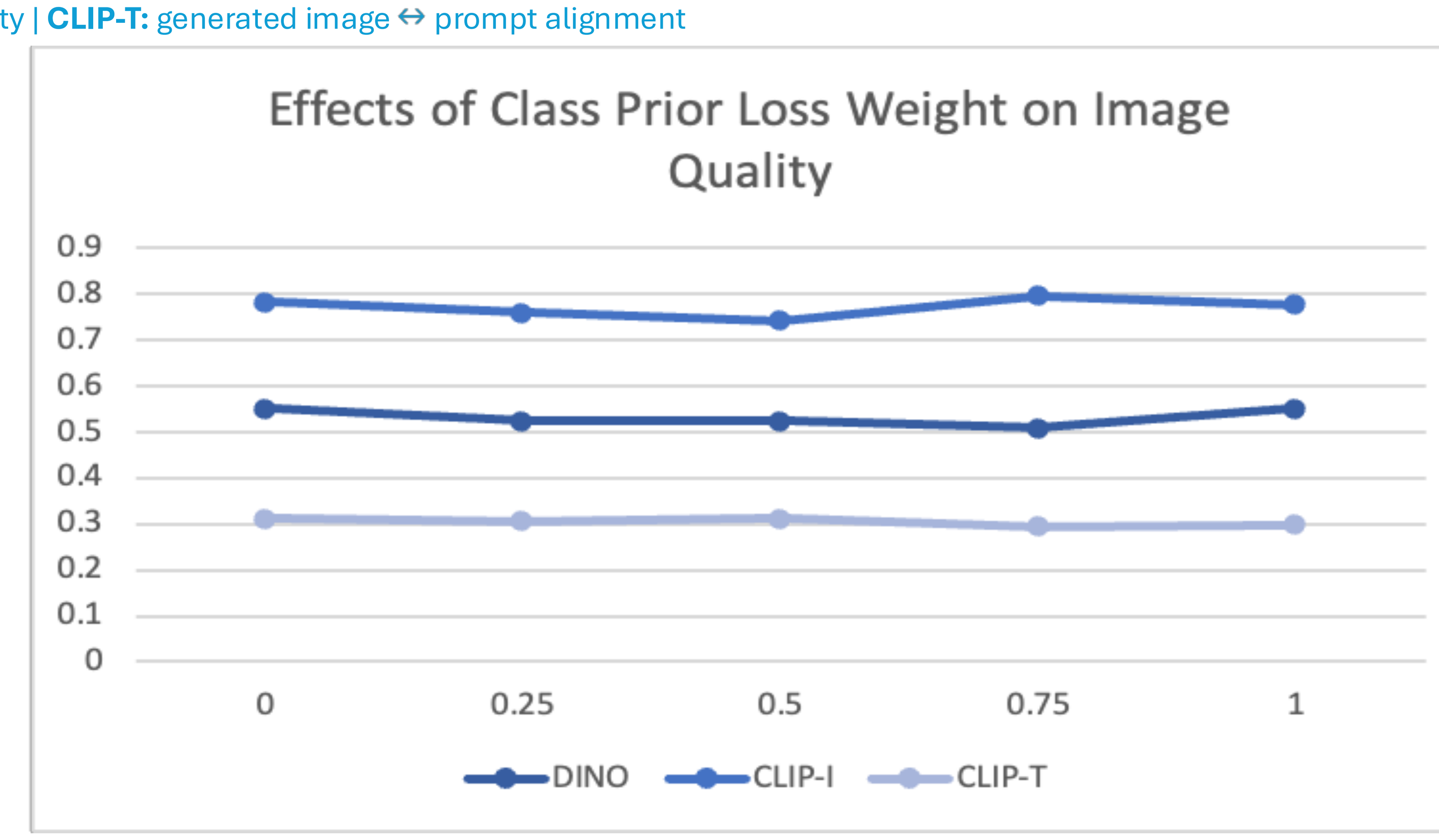


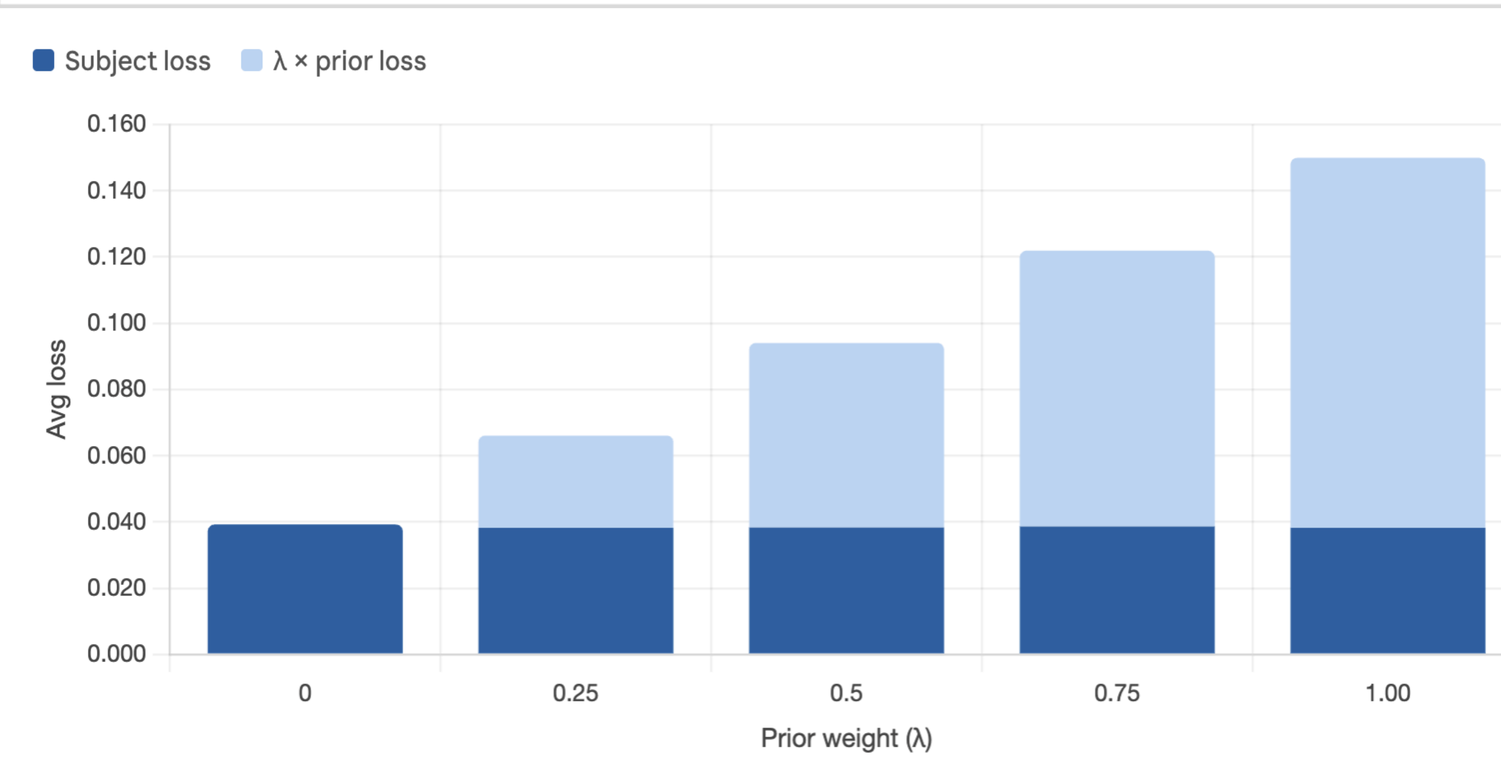
Results

A photo of a [V] water bottle on a beach

	DINO	CLIP-I	CLIP-T
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (SD)	0.668	0.803	0.305
Reimplementation (SD)	0.554	0.826	0.313

DINO: generated image ↔ reference image subject similarity | **CLIP-I:** generated image ↔ reference image overall image similarity | **CLIP-T:** generated image ↔ prompt alignment





Subject learning (how well the model learned our specific water bottle) was mainly driven by the instance images and **identifier/cross-attention training**

Discussion

Model still works well **without strong prior preservation**

Takeaway: Useful for **fast, low-memory subject binding** with low or no prior preservation

Use case: Train on a **few images** of a product, pet, or personal object and recontextualize it without full DreamBooth memory cost

Tradeoff: Cheaper and faster, but **weaker at exact details** like color, markings, and shape

Conclusion

- **Successful generations:**
  - High prompt adherence
  - Identity preservation across different contexts
  - Partially succeed in binding the unique identifier [V] with the target subject due to memory constraints
- Distinct "AI-generated" artifacts and some unnatural aspects
- Next step: test LoRA-based Dreambooth

References

[1] Ruiz, Nataniel et al. “DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation.” *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2022): 22500-22510.